

160,000–45,000 YA

45,000–35,000 YA

35,000–28,000 YA

These cave paintings from Lascaux, France, date to around 17,000 years ago. Most cave paintings are from a similar period, though some were created by the earliest *Homo sapiens* to arrive in western Europe, around 32,000 years ago.

IN AFRICA, HOMININ FOSSILS

gradually began to reveal the characteristic skeletal traits of *Homo sapiens* from around 400,000 YA: smaller brow ridges, higher and rounder skulls, and chins. DNA analysis of living humans suggests that the **common ancestor of all living humans** (known as Mitochondrial Eve) lived in Africa around 200,000 YA. An Ethiopian fossil



Prepared core and flake
Neanderthals and other hominins prepared a stone core before striking off a sharp flake to use. In Europe this technology is known as the "Mousterian."

250,000 YEARS AGO WHEN HOMO SAPIENS FIRST APPEARED

skull from 160,000 YA is almost modern in shape; this has been identified as a subspecies of modern humans, *Homo sapiens idaltu*. Humans moved north into Western Asia some time before 100,000 YA, but they do not seem to have stayed there for long.

It is debated whether uniquely human behaviors such as **language and the ability to use symbols** evolved before or after modern human anatomy. One theory is that such behaviors became vital only after 74,000 YA, when the massive eruption of **Mount Toba** in Indonesia triggered a global "volcanic winter." DNA analysis suggests that many human groups died out at this

time and, in such harsh conditions, complex modern language and symbolism would have allowed groups to **exchange resources and information** with one another, which could have made the difference between survival and extinction. However, others argue that the impact of the eruption of Mount Toba has been exaggerated, and that archaeology in Africa suggests complex hunting practices and the development of symbolism even before this.

It is not clear when **modern humans first spread into Eurasia**. Some researchers argue they left Arabia before 74,000 YA. Others say the major migration occurred later, 50,000 YA, and via western Asia, after developing a new form of stone-tool technology that involved producing long, thin flint "blades," which probably formed part of composite tools.

HUMANS SPREAD RAPIDLY

across Europe and Asia. In **Europe**, modern humans appeared in Turkey from 40,000 YA, and in western Europe shortly afterward. In Asia, fossils of *Homo sapiens* in **Indonesia** and **China** date to at least 42,000 YA, and the sea crossing to **Australia** occurred before 45,000 YA. These dates suggest that the earliest modern humans in Asia may have encountered groups of *Homo erectus*, who survived in China until at least 40,000 years ago. In Indonesia the picture was even more complicated. Fossils found on the **island of**

Flores date to less than 38,000 years ago, and seem to represent specialized, extremely small forms of *Homo erectus*, or perhaps even the descendants of earlier hominins. More evidence comes from **Denisova Cave** in Russia—DNA analysis of bones found here reveals **genetic material** distinct from that of both modern humans and Neanderthals, dated to around 40,000 YA. It seems increasingly likely that several groups descended from hominins who left Africa before *Homo sapiens* may have **coexisted in Eurasia** at this time.



HOMO SAPIENS



NEANDERTHAL

MODERN HUMANS AND NEANDERTHALS

Neanderthal skulls (right) were about the same size as anatomically modern human skulls (left), but they had lower, more sloping foreheads and a double arch of bone over their eyes that created heavy brow ridges. Their lower faces jutted out and they did not have chins. Overall, Neanderthal skeletons reveal that they were much more muscular than modern humans, as well as being extremely physically active and well-adapted to cold climates.

“THE NEANDERTHALS WERE NOT APE-MEN... THEY WERE AS HUMAN AS US, BUT THEY REPRESENTED A DIFFERENT BRAND OF HUMANITY.”

Chris Stringer and Clive Gamble, from *In Search of the Neanderthals*, 1993

IN EUROPE, MODERN HUMANS

overlapped with Neanderthals, who survived until at least 30,000 years ago. How and why **Neanderthals died out** is one of the most intensely debated topics in archaeology. There is little evidence of violent interactions between the species, and comparison of DNA increasingly suggests that there may have been some exchange of **mating partners** between the groups.

Early humans may have **outcompeted** their relatives for **food and raw materials** in the rapidly changing environmental conditions. Environments at the time were highly unstable, so even a slight increase in competition could have been significant. However, populations were small and spread out, and coexisted for up to 10,000 years in Europe, and more than 30,000

